

2 **Goodness-of-fit for multi-distribution neutrino** 3 **cross-section measurements with shared events**

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7 **ABSTRACT:** When multiple differential cross-section measurements are extracted from a common
8 event sample, the event-sharing structure between distributions imposes exact linear constraints
9 among the bin counts that reduce the effective dimensionality of the measurement below the total bin
10 count. When the statistical covariance matrix accounts for these inter-distribution correlations, the
11 constraints manifest as a rank deficiency. Systematic uncertainties may formally restore invertibility,
12 but the resulting near-singular directions yield unphysically inflated χ^2 values in global goodness-
13 of-fit tests. We present the range-projected χ^2 , a test statistic that restricts the goodness-of-fit
14 test to the subspace carrying independent statistical information, yielding a χ^2 with $N_{\text{bins}} - N_{\text{null}}$
15 degrees of freedom, where N_{null} is the number of independent constraints. We derive N_{null} from
16 the bin-combination structure of the measurement, and show how it decomposes into structural
17 contributions determined a priori from the binning geometry and kinematic contributions that
18 depend on the phase-space occupancy. The method is validated with an analytical toy model and a
19 realistic simulated neutrino–argon cross-section measurement including unfolding and multi-source
20 systematic uncertainties.

21 **KEYWORDS:** Analysis and statistical methods

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45 1 Introduction

46 Modern neutrino cross-section literature is experiencing an increase of published measurements
47 reporting multiple differential distributions extracted from the same event sample. MicroBooNE
48 has reported simultaneous measurements of cross sections for final states with and without protons
49 [1, 2], multi-differential QE-like cross section measurements in kinematic imbalance variables
50 [3, 4], and triple-differential inclusive results [5]. Other relevant examples include the multiple
51 final-state kinematic projections from T2K [6, 7], and double-differential inclusive results from
52 NOvA [8–10]. In each of these analyses, the same selected events populate bins in multiple
53 reported distributions, leading to strong inter-measurement statistical correlations. Separate but

related efforts have produced simultaneous measurements across different nuclear targets [11–13] and combined neutrino-antineutrino results [14], where correlations arise from shared systematic uncertainties rather than shared events.

When multiple distributions are extracted from a common event sample, the event-sharing structure imposes exact linear constraints among the bin counts. If every selected event populates one bin in each distribution, then the total number of events in all of them is exactly identical. More generally, the pattern of shared bin occupancy across distributions determines a set of constraints that reduce the effective dimensionality of the measurement below the total bin count. These constraints are a property of the measurement itself, independent of any particular statistical treatment.

The blockwise unfolding framework of Ref. [15] provides a rigorous method for computing the full cross-distribution covariance matrices in such measurements. In this approach, all bins corresponding to the same measured observable in the multi-distribution analysis are organized in blocks, each corresponding to a single differential cross-section distribution. Each bin block is unfolded independently using any standard method (e.g. iterative D’Agostini unfolding [16] or Wiener-SVD [17]), while the full covariance matrix, encoding the correlations between bins from different distributions, is constructed and propagated through the cross section extraction procedure. The statistical covariance for bins in different blocks is determined by the number of events they share. The constraints described above can manifest as a rank deficiency in the statistical covariance matrix, as certain directions in bin space carry identically zero variance. The number of these degenerate directions is exactly predictable from the binning geometry.

A natural use of these combined measurements is computing global goodness-of-fit metrics, such as a χ^2 statistic, that test a model against all distributions simultaneously. Compared to evaluating each distribution independently, a global test that accounts for all inter-block correlations provides the maximum discriminating power available from the data. Such global metrics are essential for interaction model tuning efforts [18–20], where measurements of multiple kinematic distributions provide complementary sensitivity to different aspects of the underlying physics. However, a global χ^2 that does not account for the constrained directions yields a test with the wrong effective number of degrees of freedom. When the statistical covariance matrix is constructed under the Poisson assumption, the constraints appear as exact null directions that prevent direct matrix inversion. Systematic uncertainties may formally restore invertibility, but the resulting near-singular directions produce unphysically inflated χ^2 values. In either case, the test is unreliable unless the constrained directions are properly identified and excluded.

We present a method to correctly compute a global goodness-of-fit test statistic for multi-distribution measurements with shared events. The approach is based on identifying the constrained directions from the bin structure of the measurement and projecting the statistic onto the subspace where independent statistical information is available. We refer to the resulting test statistic as the range-projected χ^2 . We demonstrate the method using a pedagogical toy example and connect it to the established statistical literature on hypothesis testing with singular covariance matrices.

2 Constraints from shared events

Consider a neutrino cross-section analysis simultaneously measuring B different flux-integrated differential cross sections, each with N_b true bins. If the measurement is extracted from a common

event sample then each selected event will populate exactly one bin in each distribution it belongs to. Recent MicroBooNE analyses [21] have adopted the blockwise unfolding approach presented in Ref. [15] to simultaneous cross-section measurements.

This method provides a systematic framework for reporting the correlations between the different extracted differential cross sections. It provides a general prescription to construct statistical covariance matrices that respect the correlations between bins in different distributions or blocks. The full inter-block covariance matrix constructed in reconstructed space is propagated through unfolding using a block-diagonal unfolding matrix, obtained as a direct sum of the individual unfolding matrices computed for each block. The result of the combined measurement is a vector of $N_{\text{bins}} = \sum_{b=1}^B N_b$ cross-section values with an $N_{\text{bins}} \times N_{\text{bins}}$ covariance matrix. The total covariance decomposes as $C_{\text{total}} = C_{\text{stat}} + C_{\text{syst}}$, where the statistical covariance component, computed from the shared-event structure, encodes the constraints described in the introduction.

2.1 Statistical covariance from shared events

The core idea behind the treatment of the statistical correlations in the blockwise unfolding approach is the partition of any arbitrary pair of overlapping kinematic bins X and Y into a set of three non-overlapping bins u , v , and w . The events in the intersection $X \cap Y$ are assigned to bin v , while bin u contains the events in the relative complement of Y in X (i.e. $X \setminus Y$), and bin w corresponds to the relative complement of X in Y (i.e. $Y \setminus X$). Denoting the number of predicted events in each bin as n_i , and assuming these follow independent Poisson distributions, the covariance between the counts in bins X and Y is given by [15]:

$$\begin{aligned} \text{Cov}(n_X, n_Y) &= \text{Cov}(n_u + n_v, n_v + n_w) \\ &= \text{Cov}(n_u, n_v) + \text{Cov}(n_u, n_w) + \text{Cov}(n_v, n_v) + \text{Cov}(n_v, n_w) \\ &= \text{Var}(n_v), \end{aligned} \quad (2.1)$$

as by definition bins u , v , and w have no events in common. This implies that the covariance between any two bins can be estimated from the variance of the events they share.

The variance for a set of Monte Carlo (MC) events can be estimated from the sum of their squared weights. This gives us a general prescription to calculate the elements of the statistical covariance matrix as:

$$(C_{\text{stat}})_{ij} = \sum_{e \in (i \cap j)} w_e^2, \quad (2.2)$$

where w_e is the weight associated to event e belonging to the intersection of bins i and j . In the case of measured data the weights are 1, and the covariance element simplifies to the number of events shared between the bins n_{ij} .

After propagation through the block-diagonal unfolding, the statistical covariance matrix in unfolded space inherits the same rank structure, as the unfolding matrices for each individual block are invertible. This rank structure has a direct interpretation in terms of the number of distinct event-sharing patterns across distributions, which we discuss next.

2.2 Combination matrix and null space

The previous discussion shows that the statistical covariance between any two bins is determined solely by the events they share. We now formalize this observation to derive an exact formula for

the rank of C_{stat} , and therefore for the number of directions in bin space that carry zero statistical variance.

For each event e , we can define the indicator vector $\phi_e := (\phi_{e,i})_{i=1}^{N_{\text{bins}}}$ such that $\phi_{e,i} = 1$ if $i \in \text{bins}(e)$ and $\phi_{e,i} = 0$ if $i \notin \text{bins}(e)$, where $\text{bins}(e)$ is the set of bins event e populates across all distributions. The indicator vector acts as an identifier of each event in the combined measurement. For example, in a two-block measurement with 3 bins each, an event populating bin 2 of block 1 and bin 1 of block 2 has indicator vector $\phi_e = (0, 1, 0, 1, 0, 0)$. The statistical covariance matrix can then be re-written in terms of indicator vectors as:

$$C_{\text{stat}} = \sum_e w_e^2 \phi_e \phi_e^T. \quad (2.3)$$

This outer-product decomposition makes the rank structure of C_{stat} clear. The rank of the statistical covariance matrix equals the dimension of $\text{span}\{\phi_e\}$, which depends only on the distinct indicator vectors present, not on their weights. Since all weights are $w_e > 0$ no indicator vector is removed from the sum, and the column space of C_{stat} is exactly $\text{span}\{\phi_e\}$.

We define the bin-combination matrix Ω as the matrix whose rows are the unique indicator vectors among $\{\phi_e\}$. Each row of Ω represent a distinct pattern of bin occupancy across the blocks. By construction the row space of Ω equals $\text{span}\{\phi_e\}$, and therefore:

$$\text{rank}(C_{\text{stat}}) = \text{rank}(\Omega). \quad (2.4)$$

The number of degenerate directions is then:

$$N_{\text{null}} = N_{\text{bins}} - \text{rank}(\Omega). \quad (2.5)$$

Physically, each of the N_{null} directions corresponds to a linear combination of bins along which the measurement has identically zero variance. Any component of the residual vector along such a direction is not perceived by the statistical covariance; it is a direction that the measurement has no statistical power to constrain. Importantly, both Ω and N_{null} are directly computable from the MC event-to-bin mapping alone, before constructing any covariance matrix, confirming that the constraints are a property of the measurement geometry rather than the statistical treatment.

A vector $\mathbf{x}$ is in the null space of C_{stat} if and only if, for every event e :

$$\sum_{i \in \text{bins}(e)} x_i = 0, \quad (2.6)$$

or equivalently $\Omega \mathbf{x} = 0$. Each unique indicator vector acts as a linear constraint on the null-space vectors, requiring that the components of $\mathbf{x}$ sum to zero over the bins populated by the different events. The null space directions can be divided into two physically distinct categories: structural constraints that follow from the block topology alone, and kinematic constraints that depend on which regions of phase space are populated.

2.3 Determination of the structural null space

The combination matrix Ω presented previously is constructed from MC events. One determines which events populate the different bins and collects the unique indicator vectors. However, the

indicator vectors that can appear are constrained by two distinct mechanisms. The first is purely structural, the geometry of the binning scheme forbids certain bin combinations. The second is kinematic, correlations between the different variables render some combinations unreachable even though they are geometrically allowed. These constraints are imposed by the underlying physics effects, mainly the differential cross section and the detector acceptance. In the following, we show that the structural constraints can be determined a priori, without generating MC events, from the bin edges used.

The variables used in a measurement are drawn from a set $\{v_1, \dots, v_D\}$. Each block b uses a subset of these variables, $\text{vars}(b) \subseteq \{v_1, \dots, v_D\}$. Restricting the discussion to combinations of single- and double-differential measurements, we have $|\text{vars}(b)| \in \{1, 2\}$ for every b . For each variable v used in a given block b , the binning scheme defines a set of bin edges $E_v^{(b)}$ that partition the range of v into intervals. In the case of a two-dimensional block, the edges in the binning variable may be slice-dependent, but nevertheless these can be collected into a single set of edges per variable across slices. For each variable $v \in \bigcup_b \text{vars}(b)$, we can define the refined edges:

$$\mathcal{E}_v = \bigcup_{b: v \in \text{vars}(b)} E_v^{(b)}, \quad (2.7)$$

with $M_v = |\mathcal{E}_v| - 1$ denoting the number of refined bins for variable v . Since $\mathcal{E}_v$ contains all the bin edges from every block that uses that variable, each refined bin is contained entirely within one original bin of every such block.

The common refinement of all blocks is the Cartesian product of the refined bins across all variables, denoted by $\mathcal{R}$. Each cell $c \in \mathcal{R}$ specifies a refined bin index for every variable in the measurement. For every block b , there exists a unique mapping $\beta_b(c)$ between each cell c and an original bin in said block. The elements of the indicator vector $\phi_c := (\phi_{c,i})_{i=1}^{N_{\text{bins}}}$ associated to cell c can be written as:

$$\phi_{c,i} = \begin{cases} 1 & \text{if } i = \beta_b(c) \text{ for any block } b, \\ 0 & \text{otherwise.} \end{cases} \quad (2.8)$$

If block b does not use variable v then β_b is independent of the refined bin index in v . Therefore, multiple cells may produce identical indicator vectors.

The structural combination matrix Ω_{struct} is defined as the matrix whose rows are the unique indicator vectors from the cells of the common refinement $\{\phi_c : c \in \mathcal{R}\}$. By construction, every indicator vector that appears in the Ω matrix obtained from the MC events is also present in Ω_{struct} , therefore:

$$\text{rank}(\Omega) \leq \text{rank}(\Omega_{\text{struct}}) \leq N_{\text{bins}}. \quad (2.9)$$

This hierarchy motivates a decomposition of the total number of degenerate directions:

$$\begin{aligned} N_{\text{null}} &= N_{\text{bins}} - \text{rank}(\Omega_{\text{struct}}) + \text{rank}(\Omega_{\text{struct}}) - \text{rank}(\Omega) \\ &= N_{\text{null}}^{\text{struct}} + N_{\text{null}}^{\text{kinem}}. \end{aligned} \quad (2.10)$$

The first term, $N_{\text{null}}^{\text{struct}}$, counts null directions that arise purely from the binning geometry and is computable a priori from the bin edges alone. The second term, $N_{\text{null}}^{\text{kinem}}$, accounts for additional null directions introduced by phase-space restrictions that exclude geometrically allowed combinations

and can only be determined from Monte Carlo. A description of the algorithm used to compute Ω_{struct} is presented in Appendix A.

The presence of kinematic null directions indicate that certain regions of the combined bin space are empty, which may signal that the binning extends beyond the kinematic reach of the measurement or that it is finer than the available statistics can support. In such cases, merging or removing the offending bins may be preferable to relying on a projection to handle the resulting constraints. The structural null directions, by contrast are an unavoidable consequence of measuring multiple distributions from shared events and cannot be eliminated by rebinning.

The structure of the null space of Ω_{struct} depends on how the bin edges for the different blocks overlap. In the simplest case, when blocks do not share variables, every combination of one bin per block is allowed. The resulting null space is $(B - 1)$ -dimensional, with a single constraint that requires all block totals to be equal. When blocks share one or more variables, the allowed combinations are restricted and the dimension of the null space can be different from $B - 1$. The number of null directions depends on how shared variable bin edges are aligned between blocks. Aligned slice edges decouple the measurement into different sub-problems, adding additional sum rules and increasing $N_{\text{null}}^{\text{struct}}$. On the other hand, misaligned edges couple adjacent distributions, potentially reducing the number of constraints. In the general case, the interplay between shared variables, bin edge alignment, and block topology makes a closed-form expression impractical, but the computation of $\text{rank}(\Omega_{\text{struct}})$ from the bin edge specifications is straightforward.

The structural and kinematic components of the null space respond differently to systematic uncertainties. Whether these cause the rank deficiency of C_{stat} to persist, partially resolve, or fully resolve in the total covariance C_{total} depends on the nature of the systematic treatment, which we discuss next.

2.4 Effect of systematic uncertainties

Previously, we established that the event-sharing structure imposes $N_{\text{null}} = N_{\text{bins}} - \text{rank}(\Omega)$ exact linear constraints on the bin counts, which manifest as a rank deficiency of C_{stat} . Whether the total covariance matrix inherits the rank deficiency depends on which of these null directions receive support from the systematic uncertainties. Now, we examine the mechanisms by which different systematic treatments do or do not populate the null space of C_{stat} .

A systematic variation s produces a modified vector of expected events $\mathbf{n}_s$, whose departure from the central-value prediction can be expressed in terms of a shift vector:

$$\delta_s = \mathbf{n}_s - \mathbf{n}_{\text{CV}}. \quad (2.11)$$

The different contributions to C_{syst} are constructed from the outer product of these shift vectors. A null direction $\mathbf{x}$ of C_{stat} remains a null direction of C_{total} if and only if $\mathbf{x} \cdot \delta_s = 0$ for every systematic variation in the uncertainty budget. The question is therefore whether the shift vectors have nonzero projections onto the structural and kinematic null spaces.

Consider first the simplest case, where a systematic s acts by reweighting the reconstructed event counts directly. Each event e is assigned a modified weight $w_e^{(s)}$ while preserving its bin assignments. The shift vector can then be written as:

$$\delta_s = \sum_e (w_e^{(s)} - w_e) \phi_e. \quad (2.12)$$

For any null vector $\mathbf{x}$ which satisfies the null-space condition of Eq. (2.6), the projection of $\mathbf{x}$ onto the shift vector gives:

$$\begin{aligned}\mathbf{x} \cdot \boldsymbol{\delta}_s &= \sum_e (w_e^{(s)} - w_e) (\boldsymbol{\phi}_e \cdot \mathbf{x}) \\ &= \sum_e (w_e^{(s)} - w_e) \sum_{i=1}^{N_{\text{bins}}} \phi_{e,i} x_i \\ &= 0,\end{aligned}\tag{2.13}$$

since the inner sum vanishes for every event. Direct reweighting therefore preserves the entire null space, adding no rank to C_{total} . Flux uncertainties, which modify the neutrino energy spectrum through per-event weights, are the classic example. Cross-section model variations and hadronic reinteraction uncertainties, when implemented as per-event reweighting, behave identically.

A qualitatively different situation arises when the systematic uncertainties are propagated through the response matrix rather than changing the event counts per bin directly. In this approach, adopted as the standard prescription in MicroBooNE for neutrino interaction uncertainties [15], a modified response matrix Δ'_s , mapping true bins to reconstructed bins under the given variation, is constructed for each systematic uncertainty. This matrix is obtained either by reweighting the events used to build the nominal response matrix or from an independent MC sample with varied simulation parameters. In this case, the predicted reconstructed counts for systematic variation s are:

$$\mathbf{n}_s = \Delta'_s \boldsymbol{\varphi}_{\text{CV}},\tag{2.14}$$

where $\boldsymbol{\varphi}_{\text{CV}}$ is the central-value truth prediction. The shift vector is then:

$$\boldsymbol{\delta}_s = (\Delta'_s - \Delta) \boldsymbol{\varphi}_{\text{CV}}.\tag{2.15}$$

Since Δ is block-diagonal, the difference $\Delta'_s - \Delta$ acts independently on each block, projecting the nominal truth vector through block-specific changes in efficiency and migration. The per-event coupling that makes direct reweighting respect all null directions is absent in this prescription. When the weight of an event changes, it affects simultaneously every block it populates; the response-matrix method, in contrast, processes each block independently through its own sub-matrix. The projection of $\boldsymbol{\delta}_s$ onto any null vector, whether structural or kinematic, therefore involves differences in how each block responds to the same truth-level prediction, and in general is nonzero. Response-matrix variations thus break both types of null directions, regardless of how Δ'_s is obtained.

A third mechanism applies when systematic uncertainties are evaluated by directly comparing reconstructed event counts from independent MC samples simulated with nominal and varied parameters. In this case, the shift vector takes the form:

$$\boldsymbol{\delta}_s = \sum_{e'} w_{e'}^{(s)} \boldsymbol{\phi}_{e'}^{(s)} - \sum_e w_e \boldsymbol{\phi}_e,\tag{2.16}$$

where the first sum runs over the events in the varied-model sample and the second sum over the nominal sample. Since each event in both samples populates exactly one bin per block, their difference projects to zero onto the structural null directions by an analogous argument to that used for the direct reweighting. Kinematic null directions, however, can be lifted if the

variation scatters events into reconstructed-bin combinations absent in the nominal sample. In that case, new indicator vectors appear that are not rows of Ω , potentially breaking the corresponding constraints. Detector systematic uncertainties, which usually require independent samples with modified simulation parameters that cannot be captured by event reweighting, are the typical example where this prescription is applied.

In practice, whether the eigenvalues of C_{total} along the formerly null directions are exactly zero or merely smaller than the physical eigenvalues, including these directions in a χ^2 test via standard matrix inversion produces inflated values with the wrong effective number of degrees of freedom. Because the formerly null directions carry no statistical information, C_{total} assigns them a variance set entirely by the systematic model, and the inverse covariance weights them by the reciprocal of that small value. Any residual between data and prediction that projects onto these directions is then amplified, producing a contribution to the χ^2 far in excess of one per degree of freedom. A valid goodness-of-fit test must therefore be restricted to the subspace where the measurement carries genuine sensitivity, which is the role of the projection method developed in the following section.

3 Range-projected χ^2 test statistic

Before constructing the test statistic, we must establish why restricting it to the non-null subspace discards no physical information. Along a null direction $\mathbf{x}$ of C_{stat} the event-sharing constraint $\Omega\mathbf{x} = 0$ holds identically. The same selected events enter the bin combinations on both sides, so the data value along $\mathbf{x}$ is fixed by event conservation rather than measured. Any prediction, forward-folded through the detector response matrix, respects the identical constraint. The residual along $\mathbf{x}$ is therefore zero by construction, up to systematic or numerical excursions that carry no statistical meaning. Therefore, the measurement has no power to constrain a model in this direction. Removing these directions thus removes no genuine sensitivity, only the ill-defined contributions identified in Sec. 2.4. This holds exactly to the extent that the event-sharing structure is exact; when sharing is only approximate the null directions acquire small but nonzero variance, a regime we return to in Sec. ???. Under the exact-sharing assumption, then, it is safe to formulate the global goodness-of-fit test statistic in the subspace orthogonal to the constrained directions, which we now construct.

The constraints are most naturally identified in reconstructed space, where the combination matrix is constructed. The non-null subspace can be obtained decomposing $\Omega^T\Omega$. This matrix is a square $N_{\text{bins}} \times N_{\text{bins}}$ matrix with the same column space and null space as Ω itself, making it a more compact representation of the same information. We can write its eigendecomposition as:

$$\Omega^T\Omega = V S V^T, \quad (3.1)$$

where $S = \text{diag}(s_1, \dots, s_{N_{\text{bins}}})$ contains the eigenvalues in decreasing order and V is the orthogonal matrix of eigenvectors. Of these eigenvalues, $r = \text{rank}(\Omega)$ are nonzero. Partitioning $V = [V_r \mid \tilde{V}_r]$ into the columns associated with the nonzero and zero eigenvalues, the columns of V_r span the subspace of reconstructed bin space that is not constrained by the event-sharing structure. The orthogonal projector onto this subspace is:

$$P_{\text{reco}} = V_r V_r^T. \quad (3.2)$$

Note that P_{reco} is identical to the projector onto the range of C_{stat} as defined in Eq. (2.3), since C_{stat} and $\Omega^T \Omega$ share the same column space by construction. The combination matrix formulation makes explicit the fact that the projector is determined by the measurement geometry.

Cross-section measurements are typically reported in unfolded space, and so it is natural to assume this is the space where goodness-of-fit tests should be evaluated. The blockwise unfolding matrix U , which is block-diagonal and invertible, maps reconstructed-space vectors onto unfolded space. The non-null basis vectors propagate as $W_r = U V_r$. Since U is not orthogonal in general, the columns of W_r are not orthonormal, and the projector onto their span takes the general form:

$$P_{\text{true}} = W_r (W_r^T W_r)^{-1} W_r^T. \quad (3.3)$$

Since U is invertible, W_r has the same rank as V_r and the number of constrained directions is preserved. However, as shown next, the ordering of projection and unfolding matters as the commutation relation $U P_{\text{reco}} = P_{\text{true}} U$ generically does not hold. For this reason we choose to define the test statistic in reconstructed space, where the constraints originate.

In practice, the eigenvalues of $\Omega^T \Omega$ that are technically zero take small numerical values. The threshold for identifying these can be set relative to the largest eigenvalue (e.g., $s_i/s_1 < 10^{-10}$). In most cases, a clear spectral gap of several orders of magnitude separates the physical eigenvalues from the numerical noise, making the identification unambiguous.

3.1 Test statistic definition

The goodness-of-fit test is evaluated in reconstructed space, where the residual vector is given by $\delta = \mathbf{d} - \Delta \varphi_{\text{pred}}$, where $\mathbf{d}$ is the background-subtracted data, Δ is the detector response matrix, and φ_{pred} is a truth-level model prediction. The range-projected χ^2 is defined as:

$$\chi_{\text{proj}}^2 = \delta^T (P_{\text{reco}} C_{\text{total}} P_{\text{reco}})^+ \delta, \quad (3.4)$$

with the superscript $+$ denoting the Moore–Penrose pseudoinverse.

The pseudoinverse can be avoided entirely by reducing the problem to r dimensions. One can verify that the pseudoinverse of $P_{\text{reco}} C_{\text{total}} P_{\text{reco}}$ takes the closed form:

$$(P_{\text{reco}} C_{\text{total}} P_{\text{reco}})^+ = V_r (V_r^T C_{\text{total}} V_r)^{-1} V_r^T. \quad (3.5)$$

Substituting into Eq. (3.4) and defining the reduced residual $\tilde{\delta} = V_r^T \delta$ and the reduced covariance matrix $\tilde{C}_{\text{total}} = V_r^T C_{\text{total}} V_r$, the test statistic becomes:

$$\chi_{\text{proj}}^2 = \tilde{\delta}^T \tilde{C}_{\text{total}}^{-1} \tilde{\delta}, \quad (3.6)$$

which involves only a standard inversion of the $r \times r$ matrix $\tilde{C}_{\text{total}}$. Under the null hypothesis, χ_{proj}^2 follows a χ^2 distribution with $r = N_{\text{bins}} - N_{\text{null}}$ degrees of freedom.

Without the projection, the χ^2 test statistic takes the same numerical value whether computed in reconstructed or unfolded space, as the factors of the unfolding matrix cancel identically when applying the additional smearing matrix to the predictions [17]. The introduction of the projector breaks this equivalence when the projection is performed after unfolding. In unfolded-space the residual is defined as $\mathbf{r} = \hat{\varphi} - A_C \varphi_{\text{pred}}$, where $\hat{\varphi} = U \mathbf{d}$ is the unfolded background-subtracted data

and $A_C = U \Delta$ is the additional smearing matrix. Using the propagated basis $W_r = U V_r$, one can construct an analogous test statistic in unfolded space as:

$$\chi_{\text{unf}}^2 = \tilde{\mathbf{r}}^\top \tilde{V}_{\text{total}}^{-1} \tilde{\mathbf{r}}, \quad (3.7)$$

where $V_{\text{total}} = U C_{\text{total}} U^\top$ is the total covariance matrix in unfolded space, and $\tilde{\mathbf{r}} = W_r^\top \mathbf{r}$ and $\tilde{V}_{\text{total}} = W_r^\top V_{\text{total}} W_r$ are the . While this is a valid test statistic on its own, it is not numerically equal to Eq. (3.6). Expressing it in terms of reconstructed-space quantities yields:

$$\chi_{\text{unf}}^2 = \boldsymbol{\delta}^\top U^\top U V_r (V_r^\top U^\top U C_{\text{total}} U^\top U V_r)^{-1} V_r^\top U^\top U \boldsymbol{\delta}, \quad (3.8)$$

which reduces to Eq. (3.6) only when $U^\top U = I$, i.e. when the unfolding matrix is orthogonal.

The χ^2 equivalence is restored if the projection is applied before unfolding. Projecting the reconstructed-space residual and covariance first, then propagating through the unfolding matrix, gives:

$$\begin{aligned} (U P_{\text{reco}} \boldsymbol{\delta})^\top (U P_{\text{reco}} C_{\text{total}} P_{\text{reco}} U^\top)^+ (U P_{\text{reco}} \boldsymbol{\delta}) &= \tilde{\boldsymbol{\delta}}^\top W_r^\top (W_r \tilde{C}_{\text{total}} W_r^\top)^+ W_r \tilde{\boldsymbol{\delta}} \\ &= \tilde{\boldsymbol{\delta}}^\top G (G \tilde{C}_{\text{total}} G)^{-1} G \tilde{\boldsymbol{\delta}} \\ &= \chi_{\text{proj}}^2, \end{aligned} \quad (3.9)$$

where the symmetric matrix $G = W_r^\top W_r$ acts as the unfolding matrix of the reduced system, and cancels similarly to the unprojected case. In the approach of Eq. (3.8), the full residual, including the null-space components, is first rotated by U into directions that are no longer orthogonal to the range. The unfolded-space projector then picks up this null-space contamination. Projecting first in reconstructed space avoids this by removing the null directions before unfolding can mix them with the physical subspace.

For this reason, Eq. (3.6) is the recommended definition of the range-projected χ^2 .

3.2 Statistical validity

Under the null hypothesis, the residual $\mathbf{r}$ is drawn from a multivariate Gaussian distribution with mean zero and covariance V_{total} . The reduced residual $\tilde{\mathbf{r}} = W_r^\top \mathbf{r}$ is a linear transformation of a Gaussian vector, therefore Gaussian itself with mean zero and covariance $\tilde{V}_{\text{total}} = W_r^\top V_{\text{total}} W_r$. Provided that $\tilde{V}_{\text{total}}$ is positive definite, guarantees that $\tilde{\mathbf{r}}^\top \tilde{V}_{\text{total}}^{-1} \tilde{\mathbf{r}} \sim \chi^2(r)$. The positive definiteness of $\tilde{V}_{\text{total}}$ requires V_{total} to have nonzero variance along every direction in the range of the projector P_{true} , which is generically satisfied. This can be verified by checking that all eigenvalues of $\tilde{V}_{\text{total}}$ are strictly positive.

The validity of the range-projected χ^2 can also be understood from the general theory of quadratic forms of correlated normal variables. Theorem 9.2.2 of Ref. [22] establishes that if $\mathbf{Y} \sim \mathcal{N}(0, \Sigma)$ with $\text{rank}(\Sigma) = k$, then $\mathbf{Y}^\top \Sigma^+ \mathbf{Y} \sim \chi^2(k)$ for any choice of generalized inverse Σ^+ . Assuming the null hypothesis, the projected residual $P_{\text{true}} \mathbf{r}$ is normally distributed with covariance $P_{\text{true}} V_{\text{total}} P_{\text{true}}$, which has rank r by construction. The formulation of Eq. (3.4) with the Moore–Penrose pseudoinverse is a specific instance of this general result. The key distinction from purely numerical approaches to handling singular covariance matrices is that the rank is not determined by an eigenvalue threshold applied to the covariance, but is predicted exactly via the combination matrix Ω from the structure of the measurement.

3.3 Relation to alternative approaches

4 Toy example

The formalism developed previously is now illustrated with a toy model that captures the essential features of a multi-distribution neutrino cross-section measurement: a two-dimensional underlying physics process, a simple detector model, and a multi-block binning scheme in which events are shared across distributions. The toy is design to be minimal but non-trivial. A three-block structure produces a rank-deficient statistical covariance matrix, and a Feldman–Cousins study demonstrates the consequences of ignoring or correctly treating the resulting null directions.

4.1 Model and measurement setup

The toy model mimics a muon neutrino charged-current quasi-elastic-like interaction, producing a muon characterized by its momentum p_μ and scattering angle θ_μ . The joint truth-level differential cross section is parametrized as:

$$f(p_\mu, \cos \theta_\mu) = p_\mu^k e^{-k p_\mu/p_0} e^{\kappa \cos \theta_\mu} (1 + \alpha p_\mu \cos \theta_\mu), \quad (4.1)$$

where the four parameters (k, p_0, κ, α) control the shape of the momentum spectrum, the level of forward peaking, and the strength of the momentum–angle correlation, respectively. A central-value (CV) model with $(k, p_0, \kappa, \alpha) = (3.0, 0.2, 2.0, 0.3)$ is used to build the response matrix and estimate the covariance, while a “fake” model with $(k, p_0, \kappa, \alpha) = (3.3, 0.2, 1.8, 0.35)$ serves as the true underlying distribution from which pseudo-data are drawn. Truth events are generated via rejection sampling over $p_\mu \in [0.0, 1.2]$ GeV/ c and $\cos \theta_\mu \in [-1.0, 1.0]$.

A simple detector model introduces two different effects. Momentum is smeared multiplicatively, $p_{\text{reco}} = p_{\text{true}}(1 + \delta_p)$ with $\delta_p \sim \mathcal{N}(0, \sigma_p)$ and $\sigma_p = 0.05$, while the angle is smeared additively, $\theta_{\text{reco}} = \theta_{\text{true}} + \delta_\theta$ with $\delta_\theta \sim \mathcal{N}(0, \sigma_\theta)$ and $\sigma_\theta = 1^\circ$. A sigmoid efficiency function $\varepsilon(p_{\text{reco}}) = [1 + e^{-(p_{\text{reco}} - p_{\text{thr}})/w}]^{-1}$ with threshold $p_{\text{thr}} = 0.10$ GeV/ c and width $w = 0.02$ GeV/ c models the turn-on of the event acceptance at low momentum.

The analysis uses three bin blocks in reconstructed space: a p_μ marginal (5 bins), a $\cos \theta_\mu$ marginal (5 bins), and a two-dimensional distribution sliced in $\cos \theta_\mu$ (3 slices, each with 4 bins in p_μ), for a total of $N_{\text{bins}} = 22$. The bin edges of the 2D block do not align exactly with those of the 1D marginals, which is the usual situation in real analyses. A CV sample of $N_{\text{MC}} = 500,000$ truth events is generated, processed through the detector model, and used to construct the response matrix Δ . Pseudo-data samples are generated by drawing $N_{\text{data}} = 5,000$ events from the fake model.

Systematic uncertainties are estimated following the response-matrix approach described in Sec. 2.4. Interaction-model variations are obtained by reweighting the CV sample to six alternative model configurations, each producing a varied response matrix Δ_s . The shift vector $\delta_s = (\Delta_s - \Delta) \varphi$ captures the effect on the predicted reconstructed-space distribution. Detector variations are obtained by generating eight independent samples with perturbed detector parameters ($\sigma_p, \sigma_\theta, p_{\text{thr}}$, and w), each producing a new response matrix from which a shift vector is computed similarly. The total systematic covariance matrix is $C_{\text{syst}} = N_{\text{var}}^{-1} \sum_i \delta_i \delta_i^T$, scaled to data statistics. The statistical covariance is computed using Eq. (2.2), assuming unit event weights, which captures the inter-block correlations from event sharing.

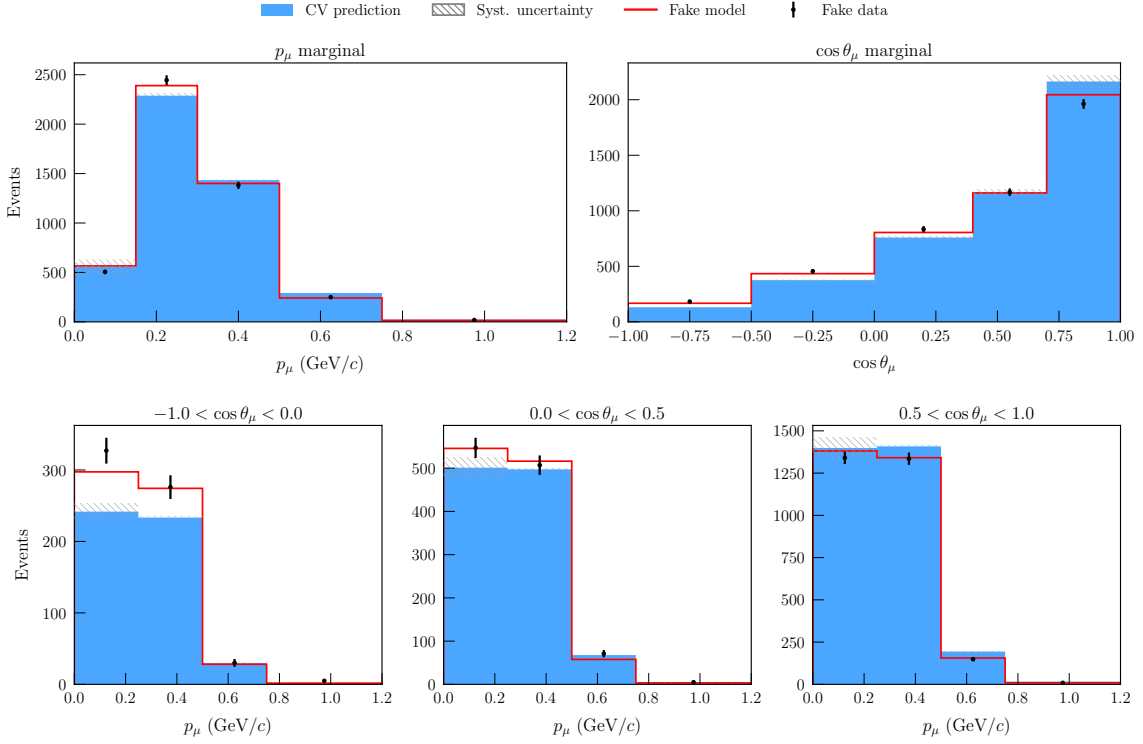


Figure 1. Predicted event distributions in the toy analysis blocks for the CV model (filled histograms) and the fake model (solid lines), together with a pseudo-data realization (points with statistical error bars). The shaded bands indicate the systematic uncertainty on the CV prediction. The top row shows the p_μ and $\cos \theta_\mu$ marginals; the bottom row shows the three $\cos \theta_\mu$ slices of the 2D block, each binned in p_μ .

Figure 1 shows the predicted event distributions for the CV and fake models in all three blocks of the toy analysis, together with a pseudo-data sample distribution and the systematic uncertainty bands.

4.2 Null-space structure

The structural combination matrix Ω_{struct} is constructed from the binning configuration of the three blocks, following the prescription from Sec. 2.3. For this bin scheme, Ω_{struct} is rank 17, yielding 5 structural null directions. The combination matrix Ω computed from the MC event list also has rank 17, confirming that all 5 null directions are structural. No additional kinematic constraints arise for this choice of physics model and binning.

The eigenvalue spectrum of C_{stat} confirms this. It shows 5 eigenvalues that are numerically zero, separated from the 17 nonzero eigenvalues by a spectral gap of several orders of magnitude. Adding systematic uncertainties constructed via the response-matrix approach lifts the 5 null eigenvalues, making $C_{\text{total}} = C_{\text{stat}} + C_{\text{syst}}$ formally full rank. However, as shown in Fig. 2, the lifted eigenvalues remain considerably smaller than the bulk of the spectrum. Including them in the test statistic introduces poorly calibrated contributions to the χ^2 , as demonstrated next.

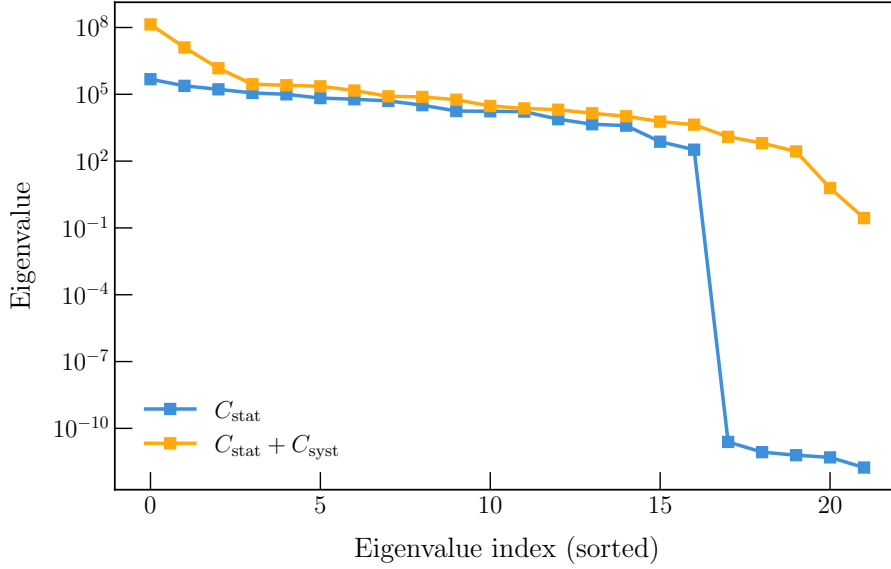


Figure 2. Eigenvalue spectra of the statistical covariance matrix C_{stat} (blue circles) and the total covariance matrix $C_{\text{total}} = C_{\text{stat}} + C_{\text{syst}}$ (orange squares), for the toy analysis, sorted in descending order.

4.3 Goodness-of-fit validation

The calibration of the projected and unprojected χ^2 test statistics is assessed with a toy Monte Carlo exercise at the true hypothesis point. A high-statistics simulation of the fake model is used to estimate the event distribution on a fine reconstructed-space grid, defined by the union of all bin edges in p_μ and $\cos \theta_\mu$ across the three blocks, corresponding to the common refinement described in Sec. 2.3. Each toy experiment draws a Poisson-fluctuated total event count and distributes them across the fine grid via multinomial sampling. These events are then projected onto the three analysis blocks, preserving the inter-block correlations from event sharing. Systematic fluctuations are added as correlated Gaussian throws from the Cholesky decomposition of C_{syst} . The χ^2 is evaluated in reconstructed space, comparing each toy to the forward-folded prediction at the true hypothesis, for both the unprojected statistic, using the direct inverse of C_{total} , and the range-projected statistic of Eq. (3.4).

Figure 3 shows the results from 10^6 toy experiments. The projected χ^2 statistic closely follows the expected $\chi^2(17)$ distribution, with a mean of 17.23 and a standard deviation of 6.01, in good agreement with the expected values of 17 and $\sqrt{2 \times 17} \approx 5.83$. In contrast, the unprojected statistic is shifted to significantly higher values with a mean of 30.59, and a standard deviation of 8.96, far above the expectation for a $\chi^2(22)$ distribution of 22 and $\sqrt{2 \times 22} \approx 6.63$. The right panel shows the corresponding p-value distributions. The projected statistic yields an approximately uniform distribution, with a false rejection rate of 5.7% at the 5% significance level. The unprojected statistic produces a false rejection rate of 32.4%, more than six times the expected value. An analyzer using the unprojected statistic for goodness-of-fit testing would falsely reject the correct model roughly one third of the time.

Writing the residual vector at the true hypothesis point as $\delta = \delta_{\text{rand}} + \mathbf{b}$, where δ_{rand} has

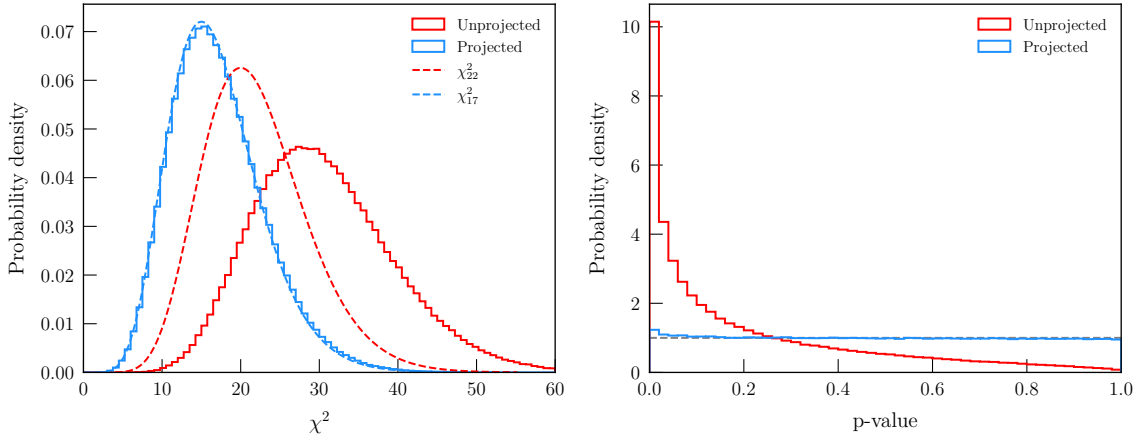


Figure 3. Goodness-of-fit validation for toy experiments generated at the true hypothesis. Left: distributions of the unprojected (red) and range-projected (blue) χ^2 statistics, compared with the expected $\chi^2(22)$ and $\chi^2(17)$ distributions (dashed curves). Right: corresponding p-value distributions, with the false rejection rates at the 5% significance level indicated.

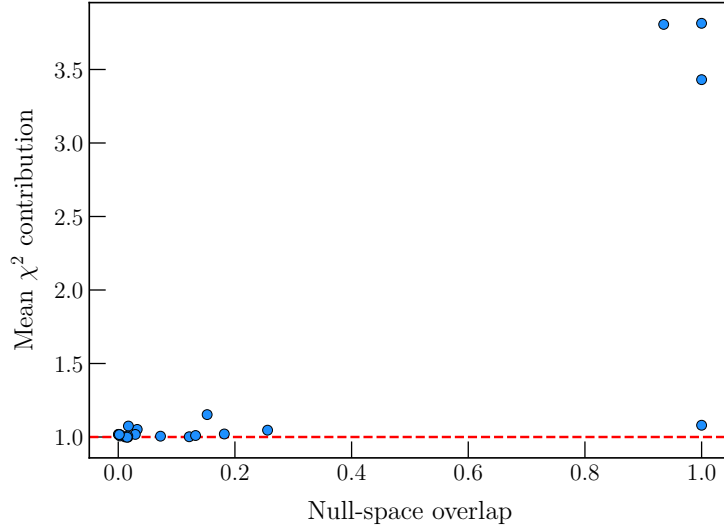


Figure 4. Mean contribution to the unprojected χ^2 from each eigendirection of C_{total} , averaged over the toy ensemble, versus the null-space overlap.

440 covariance C_{total} and $\mathbf{b} \equiv E[\boldsymbol{\delta}]$ is the deterministic data–prediction residual (estimated as the mean
 441 over the toy throws), and diagonalizing $C_{\text{total}} = \sum_j \lambda_j \mathbf{v}_j \mathbf{v}_j^T$, the unprojected statistic decomposes as:

$$\chi^2_{\text{unproj}} = \sum_j \frac{(\boldsymbol{\delta} \cdot \mathbf{v}_j)^2}{\lambda_j}, \quad (4.2)$$

442 with the expected value of each of these contributions given by:

$$E[(\boldsymbol{\delta} \cdot \mathbf{v}_j)^2 / \lambda_j] = 1 + \frac{(\mathbf{b} \cdot \mathbf{v}_j)^2}{\lambda_j}. \quad (4.3)$$

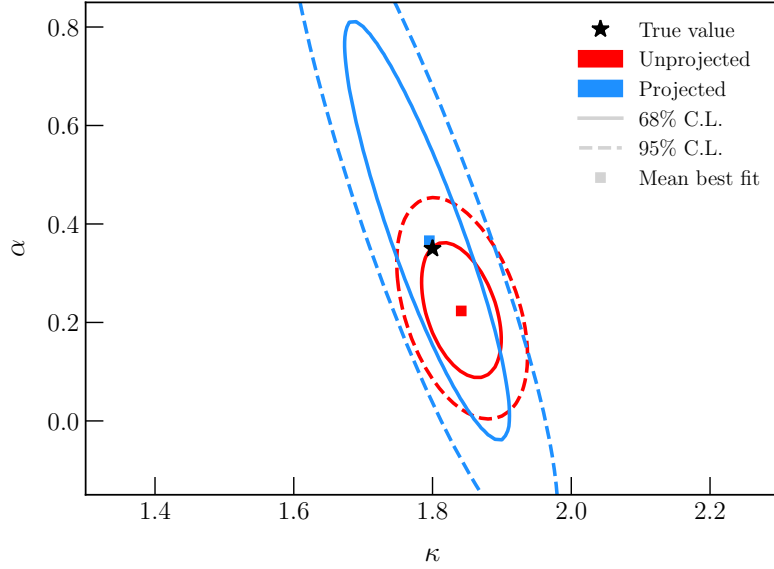


Figure 5. Confidence regions in the (κ, α) plane from the Asimov dataset, showing the 68% and 95% CL contours for the range-projected (blue) and unprojected (red) test statistics. The star marks the true parameter values; the squares indicate the mean best-fit points across 10^6 toys.

Each eigendirection contributes as a noncentral $\chi^2(1)$ with noncentrality parameter $\psi_j = (\mathbf{b} \cdot \mathbf{v}_j)^2 / \lambda_j$. On the directions overlapping the null space of C_{stat} the eigenvalue λ_j is the small lifted systematic variance, so any residual projecting onto them is amplified into a contribution far exceeding one.

To identify which directions drive the excess in the unprojected statistic, we quantify how much each eigenvector of C_{total} overlaps the null space of C_{stat} . Using the projector onto the non-null subspace P_{reco} from Eq. (3.2), we define the overlap coefficient as:

$$O_j = \mathbf{v}_j^\top (I - P_{\text{reco}}) \mathbf{v}_j, \quad (4.4)$$

which equals to one for a direction lying entirely in the null space and zero for one in the range. Figure 4 shows the mean contributions to the χ^2 test statistic across the toy ensemble against O_j . The 17 non-null directions cluster around $O_j \approx 0$ with contributions consistent with unity, confirming that the statistical fluctuation along them is correctly calibrated. The directions carrying the largest contributions are all those with substantial null-space overlap, where the small lifted eigenvalue λ_j amplifies any residual projecting onto them. Summing over all directions, the noncentrality reproduces the observed excess, $\sum_j \psi_j \approx 8.6$, accounting for the unprojected mean of 30.6.

4.4 Parameter estimation

The impact of using the projected test statistic on parameter estimation is assessed with a two-dimensional Feldman–Cousins study. A grid of hypotheses is constructed over (κ, α) with $k = 3.3$ held constant. At each grid point, the truth-level prediction is forward-folded through the nominal response matrix and compared to the toy pseudo-data. The test statistic $\Delta\chi^2 = \chi^2(\kappa, \alpha) - \chi^2_{\text{min}}$

is computed for both the unprojected and projected cases, and confidence regions are constructed from the resulting $\Delta\chi^2$ surfaces.

Figure 5 shows the 68% and 95% confidence level contours from the Asimov dataset for both test statistics. The projected contours are correctly centered on the true parameter values, while the unprojected contours are shifted, with the 68% region excluding the true point. The unprojected best fit values are biased, particularly in α . The mean best-fit value across toys is $\hat{\alpha} = 0.22$, far from the true value of 0.35, while the projected statistic recovers $\hat{\alpha} = 0.37$. The bias in κ is milder ($\hat{\kappa} = 1.84$ for unprojected, $\hat{\kappa} = 1.80$ for projected, compared to the true 1.80). This indicates that α has greater overlap with the null directions.

The frequentist coverage confirms these observations. At the 68.3% confidence level, the projected statistic achieves nominal coverage in both the joint two-dimensional region (68.2%) and the one-dimensional profiled intervals (68.1% for κ , 67.9% for α). The unprojected statistic suffers severe undercoverage: 36.5% in the joint region, 44.6% for κ , and 34.1% for α . The undercoverage is worst for α , consistent with its stronger sensitivity to the null-space contamination. The projected contours are larger than the unprojected ones, reflecting the genuine sensitivity of the measurement once the ill-constrained directions are properly removed.

5 Realistic fake data study

The toy study presented in the previous section validates the usage of the range-projected χ^2 test statistic in a controlled setting, with an analytic cross-section model and known input parameters. Now, we demonstrate that the same pathology arises, and the same fix applies, when working with a realistic measurement setup. In this case, we use a real neutrino flux, neutrino event generators, cross-section systematic uncertainties, and unfolding strategy. The latter is a key addition over the toy study, as it allows us to verify numerically the equivalence between the results in reconstructed and unfolded space, established in Sec. 3.1. In addition, we show that the projected test statistic between generator predictions,

5.1 Model and measurement setup

The central-value model is built from GENIE v3.6.2 [23] using the AR23_20i_00_000 tune on an ^{40}Ar target. The fake-data model, which plays the role of the unknown true cross section, is NuWro v25.11 [24] on the same target. Both generators are run with the SBND BNB ν_μ flux over the energy range 0–10 GeV. Since the flux is shared, flux systematic uncertainties cancel by construction and are not considered.

Events are selected using a topological $\text{CC}1\mu1p0\pi$ definition applied identically to both generators. We require exactly one muon, exactly one proton above 40 MeV kinetic energy, and no charged or neutral mesons above 50 MeV in the final state. This topological selection captures CCQE, CCMEC, and CCRES events where the pion was absorbed in final-state interactions.

The detector model is the same as in the toy study. For each candidate event, we apply a Gaussian momentum and angular smearing ($\sigma_p = 5\%$, $\sigma_\theta = 1^\circ$), and a sigmoid efficiency threshold at $p_{\text{thr}} = 100$ MeV with a width $w = 20$ MeV to the primary muon. Thus, this empty bin is merged with its neighbor, leaving $5 + 5 + (3 + 4 + 4) = 21$ bins across the three blocks.

One million GENIE events are generated and used to construct the response matrix and estimate the systematic uncertainties. One million NuWro events are generated separately; 5,000 of these are drawn as pseudo-data, with the remainder serving as the high-statistics truth prediction.

5.2 Systematic uncertainties and unfolding

Interaction-model systematic uncertainties are estimated using the GENIE Reweight framework [23], which provides per-event weights at $\pm 1\sigma$ for each systematic knob. Fifteen knobs are varied, spanning the dominant sources of uncertainty for the $CC1\mu1p0\pi$ topology: four z -expansion coefficients for the CCQE axial form factor [25], RPA screening and Pauli blocking corrections [26], the resonance axial and vector masses and overall normalization [27], the CCMEC normalization, two non-resonant single-pion background parameters, and three final-state interaction parameters controlling the pion mean free path, absorption fraction, and charge-exchange fraction [28]. Detector uncertainties are estimated similarly, using $\pm 1\sigma$ variations of the four detector parameters (σ_p , σ_θ , p_{thr} , w). The systematic covariance matrices are constructed via the response-matrix approach described in Sec. 2.4.

Each of the three distribution blocks is unfolded independently using the Wiener-SVD approach [17], with the response matrix and total data covariance computed block-by-block. The per-block unfolding matrices are assembled into the block-diagonal matrix U , and the additional smearing matrix A_C is also constructed from the per-block results. The full covariance is propagated through U as $V_{\text{total}} = U C_{\text{total}} U^T$, preserving the cross-block correlations from event sharing. Truth-level predictions are smeared with A_C before comparison to the unfolded data.

5.3 Results

The structural combination matrix Ω_{struct} , built from the three-block binning scheme, has rank 17, yielding $N_{\text{null}}^{\text{struct}} = 21 - 17 = 4$ structural null directions. The full combination matrix Ω , computed from the GENIE event list, reveals one additional kinematic null direction, bringing the total to $N_{\text{null}} = 5$. This is distinct from the bin merging mentioned in Sec. 5.1. In that case we merged a single bin of zero occupancy, whereas the kinematic null is a linear constraint among the populated bins. It arises because two cells of the common refinement corresponding to high-momentum backward-going muons are not populated by any selected events, a region where the CCQE-like cross section is strongly suppressed. Their absence removes one independent bin combination from Ω , reducing its rank to 16. The projected test statistic therefore has $r = 16$ degrees of freedom.

Figure 6 shows the reconstructed-space event distributions in all three blocks. The GENIE CV prediction and its systematic uncertainty band are compared to the NuWro fake-data realization and the NuWro prediction. The GENIE systematic band does not fully cover the NuWro prediction in several bins, reflecting genuine model differences that the χ^2 test should be able to identify.

Figure 7 shows the corresponding unfolded distributions, with the GENIE and NuWro truth-level predictions smeared through the additional smearing matrix A_C . Per-block χ^2 values, computed using only the block-diagonal covariance, are annotated in each panel. These are well-behaved for the NuWro comparison, confirming that the unfolding correctly recovers the true distribution at the per-block level. The GENIE comparison shows mild tension in some blocks.

Table 1 summarizes the global χ^2 results. The unprojected statistic yields 121.9/21 for the NuWro prediction. This results in a disproportionate rejection of the correct model, entirely

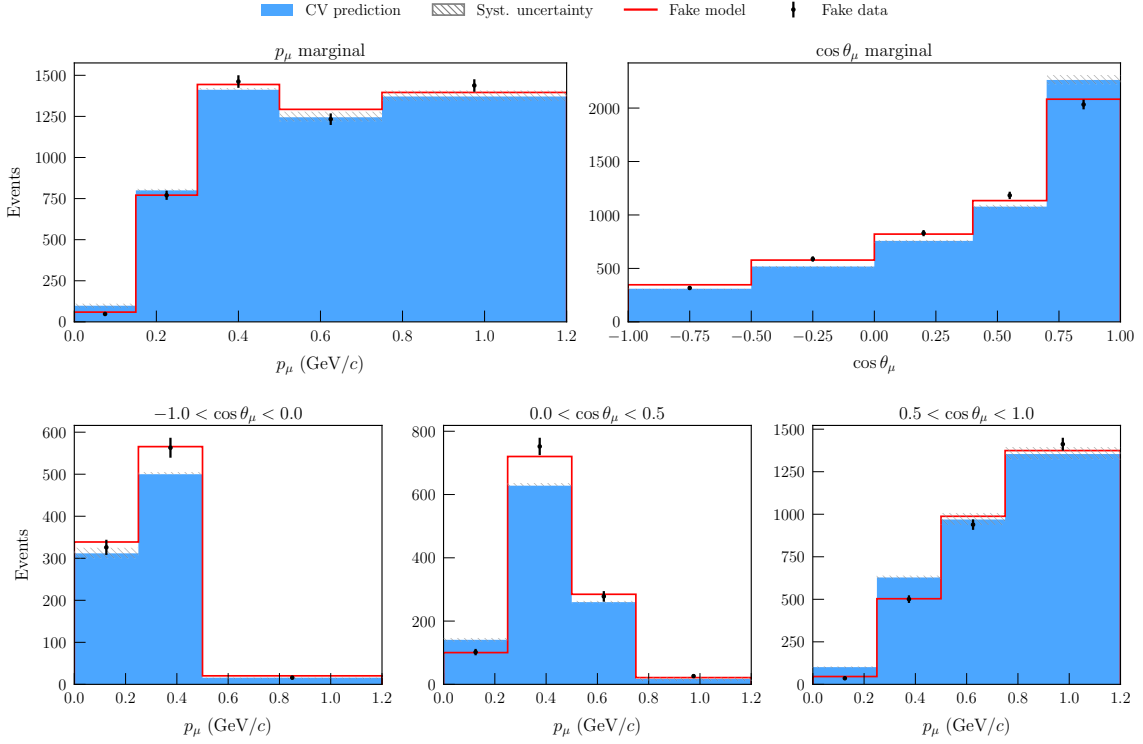


Figure 6. Reconstructed-space event distributions for the realistic fake-data study. The GENIE CV prediction (filled histograms) and systematic uncertainty bands (shaded) are compared to the NuWro prediction (red line) and a 5,000-event NuWro pseudo-data realization (points with statistical error bars). The top row shows the p_μ and $\cos \theta_\mu$ marginals; the bottom row shows the three $\cos \theta_\mu$ slices of the 2D block.

Table 1. Global χ^2 values for the NuWro fake data compared to the GENIE and NuWro truth-level predictions. The projected test statistic has $N_{\text{bins}} - N_{\text{null}} = 16$ degrees of freedom; the unprojected has $N_{\text{bins}} = 21$.

Test statistic	GENIE	NuWro
Unprojected χ^2/ndof	252.4 / 21	121.9 / 21
Projected χ^2/ndof	154.4 / 16	15.7 / 16

driven by the null-space contamination. The projected statistic gives 15.7/16, consistent with the expected $\chi^2(16)$ distribution and confirming that the method is correctly calibrated. This was further validated with 10^5 pseudo-experiments, which confirmed that the projected χ^2 distribution follows $\chi^2(16)$ while the unprojected distribution does not follow $\chi^2(21)$, consistent with the toy results of Sec. 4.3.

The projected statistic retains full discriminating power against the wrong model. The comparison to GENIE yields 154.4/16, a strong rejection indicating that the GENIE uncertainty budget cannot accommodate the NuWro prediction. By contrast, the unprojected statistic gives 252.4/21 for GENIE, also a strong rejection but indistinguishable in practice from its 121.9/21 for NuWro.

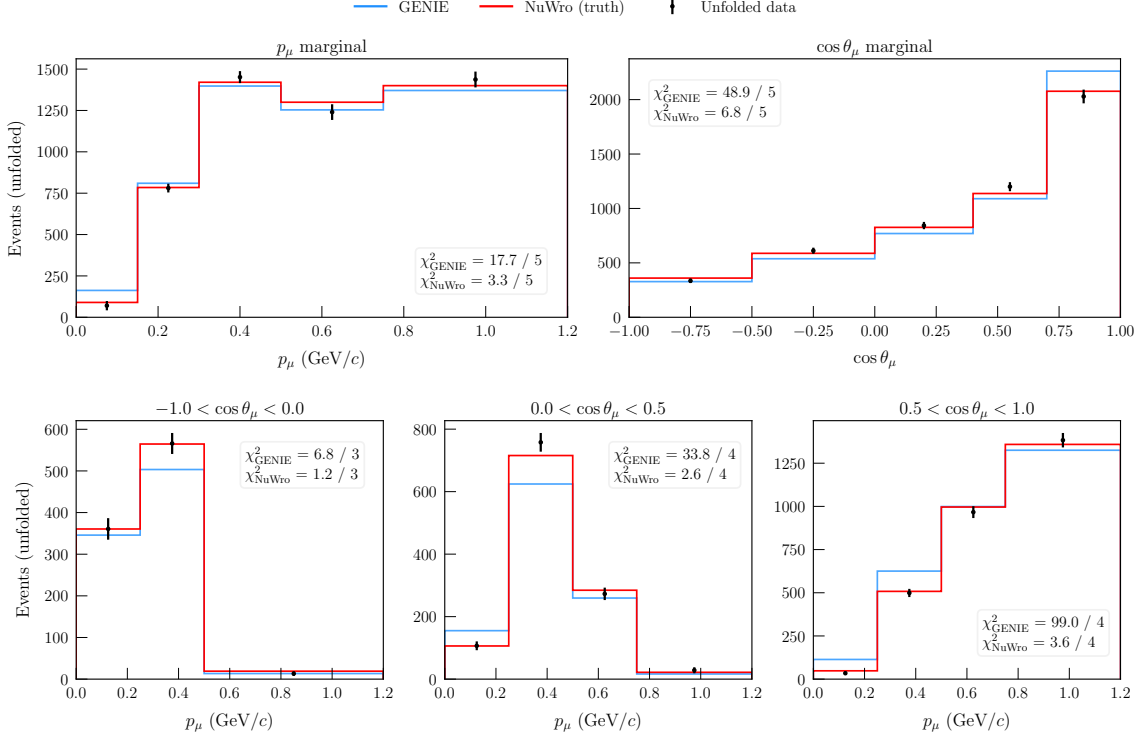


Figure 7. Unfolded event distributions for the realistic fake-data study. The GENIE (blue) and NuWro (red) truth-level predictions are smeared through the additional smearing matrix A_C for comparison with the unfolded fake data (points with propagated uncertainties). Per-block χ^2 values are annotated in each panel.

Without the projection, both the correct and incorrect models are rejected, and the test provides no useful discrimination.

Finally, the projected χ^2 was verified to give identical numerical values whether computed in reconstructed or unfolded space, for both the GENIE and NuWro predictions. This confirms the equivalence established in Sec. 3.1 and validates the practical implementation of the projection in the presence of unfolding.

6 Summary and conclusions

Acknowledgments

A Structural combination matrix algorithm

We describe the algorithm implemented to construct the structural combination matrix Ω_{struct} from the bin edge specifications of the multi-block measurement. Each row of Ω_{struct} is a unique indicator vector encoding one structurally allowed combination of bins, one per block, that a single event can simultaneously populate. The rank of this matrix determines the number of null directions, as described in Sec. 2.3.

The construction proceeds in three stages. First, the kinematic variables used across all blocks are classified as shared (appearing in two or more blocks) or private (appearing in exactly

one block). This distinction is key to the efficiency of the algorithm. Shared variables create correlations between blocks, while private variables contribute bins that are independent of all other blocks.

Second, a common refinement is computed for each shared variable by taking the union of all bin edges from every block that uses it. The algorithm then enumerates the cells of this shared-variable refinement. For each cell, the midpoint in each shared variable is used to identify which bin (or set of bins) in each block is compatible with that cell. When the binning of a given block is fully determined by the shared variables (e.g., a 1D block whose variable is shared, or a 2D block whose slice and bin variables are both shared) the cell maps to a single bin. When a block uses a private variable, the shared cell constrains only part of the binning, and the block contributes multiple candidate bins corresponding to the different private-variable choices. A 1D block whose variable is private contributes all of its bins; a 2D block with a shared slice variable and a private bin variable contributes all bins within the determined slice.

Third, the candidate bins from each block are combined. In the simplest case, the full Cartesian product of all candidate bins is expanded, and each element becomes a row of Ω_{struct} . However, when many blocks have private variables, this product can become very large. In that case, a representative subset of indicator vectors is used instead. A base combination is formed by selecting the first candidate bin from each block, and then the remaining candidates from each block are substituted one at a time while the others are held at their base values. This produces at most $1 + \sum_b (n_b - 1)$ rows per shared cell, where n_b is the number of candidate bins for block b . The representative set spans the same column space as the full Cartesian product, and therefore yields the same rank(Ω_{struct}) and null-space structure.

To understand why, note that any indicator vector from the full product can be written as the sum of indicator vectors from the representative set. Denoting the base combination as $\mathbf{e}_0$ and the single-block variations as $\mathbf{e}_j^{(k)}$ (block k switched to its j -th candidate), any full-product row that selects candidate j_k in each block satisfies:

$$\mathbf{e}_{j_1, \dots, j_B} = \mathbf{e}_0 + \sum_k (\mathbf{e}_{j_k}^{(k)} - \mathbf{e}_0), \quad (\text{A.1})$$

since the blocks occupy disjoint columns. The full product therefore lies in the span of the representative rows, and the rank is preserved.

When the measurement involves mutually exclusive event selections, blocks belonging to different selections share no events and are processed independently. The algorithm is applied separately to each selection group, and the resulting matrices are vertically concatenated.

The cost of the algorithm is dominated by the enumeration of shared-variable cells, which scales as $\prod_{v \in \text{shared}} M_v$ where M_v is the number of refined bins for shared variable v . Since only shared variables enter this product, measurements with many blocks but few shared variables, such as a set of 1D distributions in different kinematic variables, are handled efficiently regardless of the total number of bins.

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